Alibaba Cloud

Singapore customer portal

Architecture proposal

Prepared for Acme Pay Pte Ltd

Date	Revisions	Products	Catalog checked
1 November 2026	4	21	2026-10-31

Prepared with Architecture Builder

AI-generated. Review with a solutions architect before you build.

Summary

The design

I'll design a two-zone fintech customer portal in Singapore with ESA, ALB, WAF, ECS, Tair and RDS.

21 products, 22 connections, 4 revisions

Pinned requirements

Customer data must stay in Singapore	**Held** Pinned in revision 1
Customer data stays in Singapore	**Held** Pinned in revision 2

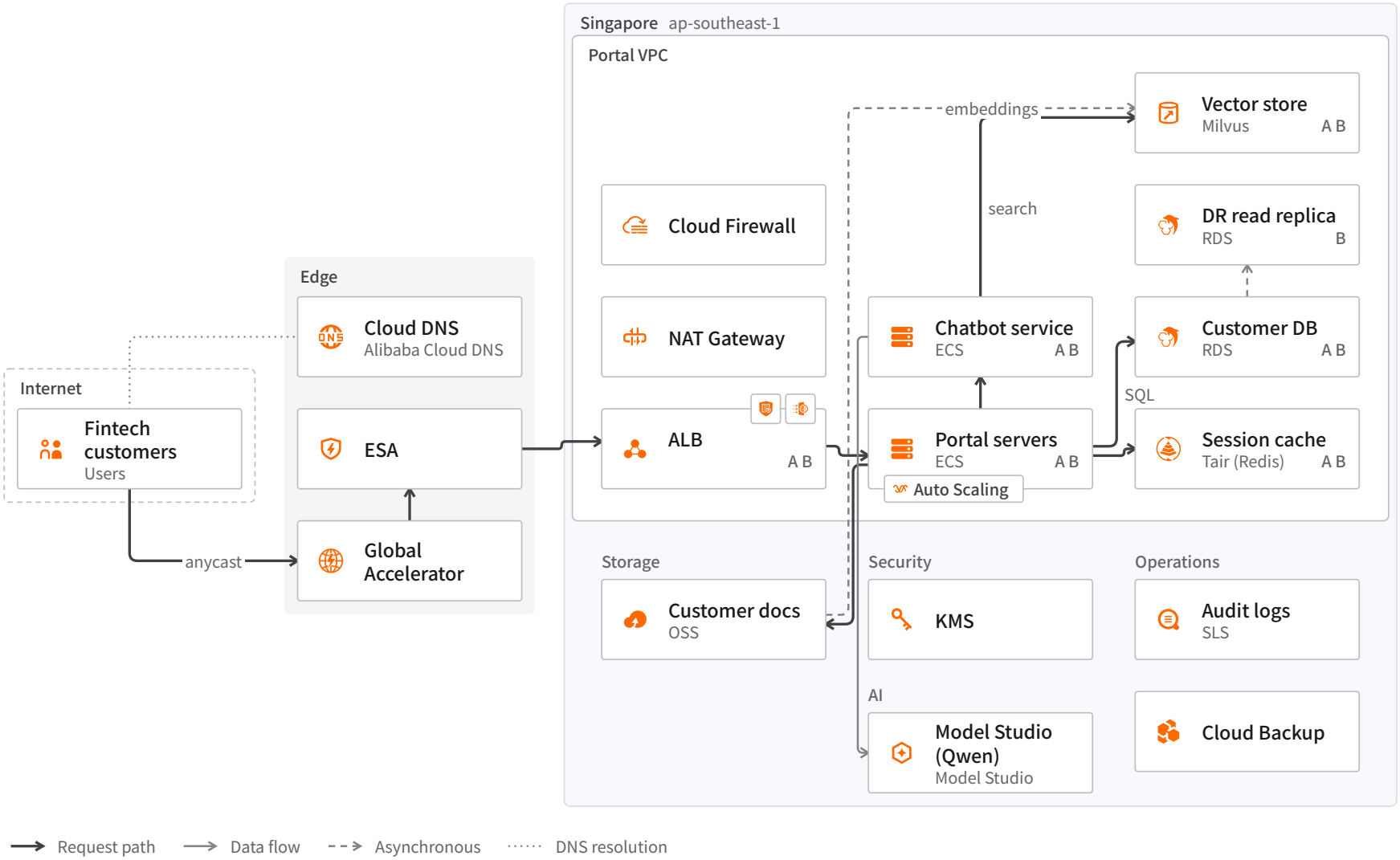
Constraints

Data and workloads stay in Singapore (ap-southeast-1)

Assumptions

No assumptions were recorded.

Architecture diagram



Components

Every product in the diagram, why it is in this design, and where to read its official documentation.

Users

Fintech customers
Internet

Role in this design
Customers accessing the portal over the internet.

What it does
People reaching the system over the internet from browsers or apps.

Official documentation
Documentation link not yet available

Alibaba Cloud DNS

Edge · Domain Names and Websites

Role in this design
Resolves the portal domain to ESA.

What it does
Authoritative public DNS for your domains; a resolve hop, not on the data path.

Official documentation
www.alibabacloud.com/product/dns

Edge Security Acceleration (ESA)

Edge · Networking & CDN

Role in this design
Accelerates portal traffic and provides DDoS defense.

What it does
Global edge: caching, dynamic acceleration, WAF, DDoS defense, bot management and edge functions.

Official documentation
www.alibabacloud.com/product/esa

Global Accelerator (GA)

Edge · Networking & CDN

Role in this design
Accelerates European user traffic to Singapore over the backbone.

What it does
Anycast entry points carrying traffic over the Alibaba backbone to regional endpoints.

Official documentation
www.alibabacloud.com/product/ga

Cloud Firewall

Singapore (ap-southeast-1) · Portal VPC · Security

Role in this design
Inspects north-south traffic into the VPC.

What it does
Internet, VPC and NAT border firewalls with intrusion prevention and access control.

Official documentation
www.alibabacloud.com/product/cfw

NAT Gateway

Singapore (ap-southeast-1) · Portal VPC · Networking & CDN

Role in this design
Outbound internet for patches without public IPs.

What it does
Gives private VPC resources outbound internet (SNAT) and mapped inbound ports (DNAT).

Official documentation
www.alibabacloud.com/product/nat

Application Load Balancer (ALB)

Singapore (ap-southeast-1) · Portal VPC · Zones A, B

Role in this design
Routes HTTPS traffic across two zones.

What it does
Layer-7 HTTP/HTTPS/gRPC load balancer with content routing; also serves as ACK Ingress.

Official documentation
www.alibabacloud.com/product/server-load-balancer

Web Application Firewall (WAF)

WAF 3.0
Singapore (ap-southeast-1) · Portal VPC · On ALB

Role in this design
Blocks OWASP attacks on the customer portal.

What it does
WAF 3.0: blocks OWASP attacks, CC floods and bots via cloud-native or CNAME access.

Official documentation
www.alibabacloud.com/product/web-application-firewall

Certificate Management Service

TLS certificate
Singapore (ap-southeast-1) · Portal VPC · On ALB

Role in this design
Terminates TLS at the load balancer.

What it does
Buy, issue and deploy SSL/TLS certificates, including a private CA.

Official documentation
www.alibabacloud.com/product/certificates

Elastic Compute Service (ECS)

Computing

Role in this design
Chatbot service: Runs the chatbot backend that orchestrates RAG queries within Singapore.
Portal servers: Runs the web app in two zones for HA.

What it does
Virtual machines (x86, Arm, GPU) with cloud disks and security groups.

Official documentation
www.alibabacloud.com/product/ecs

Auto Scaling (ESS)

Singapore (ap-southeast-1) · Portal VPC · On Portal servers

Role in this design
Adds instances during peak login times.

What it does
Scales ECS or ECI fleets by rules, schedules or metrics and registers them to load balancers.

Official documentation
www.alibabacloud.com/product/auto-scaling

Vector Retrieval Service for Milvus

Vector store
Singapore (ap-southeast-1) · Portal VPC · Zones A, B

Role in this design
Stores document embeddings for RAG retrieval, data stays in Singapore.

What it does
Managed vector database compatible with open-source Milvus, for RAG and multimodal search.

Official documentation
www.alibabacloud.com/product/milvus

ApsaraDB RDS

Database

Role in this design
DR read replica: Cross-zone read replica for disaster recovery within Singapore.

Customer DB: Primary relational store for customer data.

What it does
Managed MySQL, PostgreSQL, SQL Server and MariaDB with HA, read replicas and backups.

Official documentation
www.alibabacloud.com/product/apsaradb-rds

Tair (Redis® OSS-Compatible)

Session cache
Singapore (ap-southeast-1) · Portal VPC · Zones A, B

Role in this design
Stores sessions and tokens in memory.

What it does
Managed Redis-compatible in-memory database and cache, plus Tair enterprise features.

Official documentation
www.alibabacloud.com/product/tair

Object Storage Service (OSS)

Customer docs
Singapore (ap-southeast-1) · regional services · Storage

Role in this design
Encrypted bucket for KYC documents.

What it does
Durable object storage for files, media, backups and data lakes; S3-compatible.

Official documentation
www.alibabacloud.com/product/object-storage-service

Key Management Service (KMS)

Singapore (ap-southeast-1) · regional services · Security

Role in this design
Encrypts database and storage keys at rest.

What it does
Managed encryption keys with rotation and BYOK, plus Secrets Manager.

Official documentation
www.alibabacloud.com/product/kms

Alibaba Cloud Model Studio

Model Studio (Qwen)
Singapore (ap-southeast-1) · regional services · AI & Machine Learning

Role in this design
Provides Qwen LLM inference for generating chatbot answers.

What it does
Managed model APIs for Qwen (such as qwen3.8-max) and third-party models; OpenAI-compatible.

Price
Pay per token; qwen3.8-max $2 per 1M input and $6 per 1M output tokens (Singapore); 1M free tokens per model for 90 days, as of 2026-10-31

Official documentation
www.alibabacloud.com/product/modelstudio

Simple Log Service (SLS)

Audit logs
Singapore (ap-southeast-1) · regional services · Storage

Role in this design
Centralised audit trail for compliance.

What it does
Collects, searches, analyzes with SQL and alerts on logs, metrics and traces.

Official documentation
www.alibabacloud.com/product/log-service

Cloud Backup

Singapore (ap-southeast-1) · regional services · Storage

Role in this design
Backs up ECS, RDS and OSS for disaster recovery in Singapore.

What it does
Backup and disaster recovery for ECS, databases, NAS, OSS and VMware.

Official documentation
www.alibabacloud.com/product/hybrid-backup-recovery

Connections

How the products in the diagram talk to each other.

From	To	Carries	Type
Fintech customers	Cloud DNS		DNS resolution
ESA	ALB	origin	Data
ALB	Portal servers	HTTP	Data
Portal servers	Session cache	Redis	Data
Portal servers	Customer DB	SQL	Data
Customer DB	KMS		Control
Portal servers	NAT Gateway		Control
ALB	Cloud Firewall		Control
Portal servers	Audit logs		Control
Customer DB	Audit logs		Control
Portal servers	Customer docs	objects	Data
Customer DB	DR read replica	replication	Asynchronous
Portal servers	Cloud Backup		Control
Customer DB	Cloud Backup		Control
Customer docs	Cloud Backup		Control
Portal servers	Chatbot service	chat API	Data
Chatbot service	Vector store	search	Data
Chatbot service	Model Studio (Qwen)	LLM API	Data
Customer docs	Vector store	embeddings	Asynchronous
Chatbot service	Cloud Backup		Control
Fintech customers	Global Accelerator	anycast	Data
Global Accelerator	ESA	backbone	Data

Revision history

Every ask and what it changed in the design.

Revision 1

Drawn in 23.0 s

Ask

We're a Singapore fintech and customer data must stay in Singapore. Design a web app with a database for our customer portal.

Changes

+ Users · Fintech customers · in Internet
+ Cloud DNS
+ ESA · in Edge
+ ALB · zones A, B
+ WAF · WAF 3.0 · on ALB
+ ECS · Portal servers · zones A, B
+ Auto Scaling · on ECS
+ Tair (Redis) · Session cache · zones A, B
+ RDS · Customer DB · zones A, B
+ OSS · Customer docs · in Singapore (ap-southeast-1)
+ SLS · Audit logs · in Singapore (ap-southeast-1)
+ KMS · in Singapore (ap-southeast-1)
+ Cloud Firewall · in Portal VPC
+ NAT Gateway · in Portal VPC
+ Certificates · TLS certificate · on ALB
+ 12 connections

Revision 2

Drawn in 9.6 s

Ask

Add disaster recovery.

Changes

+ Cloud Backup · in Singapore (ap-southeast-1)
+ RDS · DR read replica · zones B
+ 4 connections

Revision 3

Drawn in 7.3 s

Ask

Add an AI chatbot that answers customer questions.

Changes

+ ECS · Chatbot service · zones A, B
+ 5 connections
+ Milvus · Vector store · zones A, B
+ Model Studio · Model Studio (Qwen) · in Singapore (ap-southeast-1)

Revision 4

Drawn in 4.5 s

Ask

Our users in Europe say the portal is slow. What can we do?

Changes

+ Global Accelerator
+ 2 connections · Fintech customers to Global Accelerator to ESA
− 1 connection removed · Fintech customers to ESA